

Medical treatments

Regular reviews play a crucial role in enhancing patient compliance with treatment, which is a key factor in the success of androgenetic alopecia (AGA) management. Since visible results may take several months, maintaining adherence to therapy is essential.

Generally, with medical treatment, a reduction in hair loss becomes noticeable after 3–6 months, and visible hair regrowth typically appears after 6–12 months. Continuous, long-term treatment is required to sustain benefits. It is important to note that current medical treatments are not curative. Patients should be clearly informed about the limitations of therapy: AGA treatment cannot restore hair to prepubertal density, and the primary goal is to halt or slow further hair loss.

Males

Two agents are approved by the US Food and Drug Administration (FDA) for the treatment of AGA in men: **topical minoxidil** and **oral finasteride**.

(1) Topical minoxidil solution

Topical minoxidil is applied as 1 mL of solution twice daily. Its exact mechanism of action is not fully understood, but it is believed to prolong the anagen (growth) phase and increase hair shaft diameter, regardless of the underlying cause of hair loss.

The efficacy of minoxidil varies across studies. It is well established that the **5% formulation** is more effective than 2% or 3% solutions. In a study of men aged 18–49 years, those using 5% minoxidil had **45% higher hair counts** compared to those using the 2% solution. A Japanese study also demonstrated the superiority of 5% over 1% minoxidil in Japanese men, and a phase 4, open-

label, multicenter trial confirmed the safety and efficacy of 5% minoxidil in Korean men with AGA.

Patients should be advised that during the first 2–8 weeks of treatment, a temporary increase in hair shedding may occur due to telogen effluvium. This is self-limiting and resolves as new anagen hairs begin to grow. It should not prompt discontinuation of treatment.

An advancement in minoxidil delivery is the **5% topical foam**, which has been shown in placebo-controlled, double-blind trials to be effective, safe, and cosmetically acceptable. The hydroalcoholic foam formulation offers convenience and improved tolerability for some patients.

(2) Oral finasteride

Finasteride is a selective inhibitor of type II 5 α -reductase, administered at a dose of **1 mg daily**. Clinical trials in men aged 18–41 years over a 2-year period reported the following outcomes:

- **At 1 year:**
 - 48% experienced hair regrowth (slight: 30%, moderate: 16%, great: 2%)
 - 51% had stabilized hair loss
 - 1% showed progressive hair loss
- **At 2 years:**
 - 66% had regrowth
 - 33% had stabilized hair loss
 - 1% continued to lose hair

The number of responsive hairs typically stabilizes after 1 year, but continued treatment improves hair length, thickness, and pigmentation, leading to better scalp coverage over time. Upon discontinuation, regrown hair is gradually lost, and the balding process resumes.

An extension of this study to 5 years confirmed that finasteride 1 mg/day is well tolerated and provides durable improvements in hair growth.

Finasteride is generally safe, with most side effects being mild and not requiring treatment cessation. Rare adverse effects include decreased libido and erectile dysfunction.

Currently, there is **no proven benefit** of finasteride in women with AGA. A placebo-controlled trial in postmenopausal Western women using 1 mg/day for 1 year showed no significant improvement. However, a study using **5 mg/day** in Asian women suggested potential benefit for normoandrogenic women with female pattern hair loss (FPHL). Further research is needed to define the role of finasteride in female patients.